The Estimator Decides:

What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards

Anonymous authors

Working draft v0.9 — August 6, 2026

Code, logs, and proofs: https://github.com/linjiw/curriculum-maxrl

Abstract

Difficulty curricula and failure recycling (hindsight relabeling) are designed today as estimator-agnostic add-ons to RL with verifiable rewards. We show they are not. *Exact result:* for success-conditioned (MaxRL-style) advantages, the expected absolute coefficient mass a prompt emits from N rollouts is exactly $2(\text{pass@}N - \text{pass@}1)$ — a compute-indexed learnability functional vanishing exactly at pass rates 0 and 1, partitioning difficulty into a dead zone, a learnable region, and a mastered tail, with the region's peak and N-scaling derived rather than tuned. *Replicated experiment:* on Countdown, exact-verifier recycling improves mean accuracy while *losing* pass@k coverage — **recycling-induced sharpening**, three seeds — and a saturation heuristic on relabel destinations returns frontier-tier coverage to baseline while keeping most of the mean gain at one validated operating point. *Practical consequence:* report pass@k coverage beside mean accuracy whenever a curriculum or recycler ships, conditioned on the estimator it ships on. *Tested against ourselves:* the sharpest coverage claim our earlier runs suggested failed its pre-registered balanced factorial at the registered endpoint and is retracted here; the time-integrated ordering that survived it was then registered on six fresh seed blocks and *confirmed* — MaxRL retains more coverage than GRPO in 6/6 paired blocks under each sampler (p=.031 per sampler; 24/24 across both waves), concentrated in the easy band the mass curves indict. We also note a refinement to MaxRL: its practical drop-all-fail estimator is unbiased for truncation order $T{=}N{-}1$, not N.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, and generation can dominate wall-clock — in our committed telemetry, generation-dominated LLM sessions average 83% of step time.[1] On hard task pools, most of that compute buys nothing: under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal** — either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field's response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [23]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator's own algebra. The organizing fact is the shape of one function. **The estimator's derived utility $u_N(p)$ — half its expected coefficient mass — vanishes exactly at pass rates 0 and 1 and peaks once in between, tracing three regimes of difficulty: a dead zone, a learnable region, and a mastered tail. Curricula reallocate compute within the learnable region; verified relabeling is a way to create a positive**

[1] Eight sessions, 282 steps; across all 1,188 logged steps including CPU-bottlenecked maze-scale sessions the step-weighted mean is 31% — the 83% figure describes the LLM regime, not every workload.

update for an all-fail requested task; and relabels that land in the mastered tail buy sharpening, not signal. The estimator shapes all three. Three questions unpack this:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected coefficient mass a task emits from a group of N rollouts has a closed form: $2(\text{pass@}N - \text{pass@}1)$, a compute-indexed learnability functional whose two exact zeros anchor the difficulty regimes (§3). Mass is a surrogate for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the learnable region's location, peak, and N-scaling come from the budget you already chose, and published learnability curricula are the $N{=}2$ slice. It also bounds the channel. Allocation can only redistribute signal that exists: a perfect-information allocator ties the full posterior-plus-recycling stack in end performance (§6.1) — so recycling compensates for allocation error rather than beating perfect allocation — and at LLM scale a prompt-level teacher starves before it pays (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N{-}1)\times$ more mass than RLOO (the unnormalized leave-one-out baseline estimator) on hard tasks as $p \to 0$ and releases mastered tasks; GRPO's variance-normalized weights keep spending on both tails. The predicted coverage ordering is robust where gradients are exact; at neural scale our first pre-registered factorial *retired its strong form* — an earlier cohort's zero-exception divergence had conflated recycling's contribution with the estimator effect — and a second factorial on fresh seed blocks then *confirmed* the time-integrated ordering that survived, 6/6 paired blocks under each sampler, concentrated where the mass curves say GRPO overspends (§6.3b tells that story once, in full). Recycling has its own failure mode, and the algebra flags it: relabels that land in the mastered tail sharpen. On Countdown arithmetic this is **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@k coverage — the data-induced sibling of GRPO's weight-induced collapse, formally hindsight's marginal-distribution shift [21] composed with curation-driven self-consumption [22].

Q3: Does the algebra also supply the fix? It motivates one. A relabel to a destination the model reliably reaches sits in the mastered tail, where mass vanishes, so we gate relabel admission by a saturation statistic on the destination: a decayed frequency estimate of how often that destination has recently appeared in the recycler's own relabel stream, thresholded before another relabel is admitted. The mastered-tail zero of the same utility that schedules sampling motivates *that such a threshold should exist*; the statistic itself is a heuristic proxy — it is not a fresh-rollout pass-rate estimate for the destination task, and the threshold value is tuned (§5, Table 3). Across three seeds the moderate gate returns frontier-tier coverage to the no-recycling baseline while keeping most of recycling's mean gain; whether gate strength traces a monotone trade is an open question §6.9 states precisely. Where recycling is healthy the gate is transparent: hard-destination relabels pass untouched, so the exact-gradient testbeds of §6.1 reproduce their results with the gate enabled. The gate is one mechanism across domains, but not switch-free: each domain supplies a relabel map and a destination key, and Jugs showed the key choice is load-bearing (§8).

Our contributions:

1. **A closed-form curriculum** (§3): the expected coefficient mass of a prompt under success-conditioned RL is exactly $2(\text{pass@}N - \text{pass@}1)$, a compute-indexed learnability functional whose peak location and N-scaling are derived rather than tuned, and whose exact zeros at $p \in \{0, 1\}$ anchor the dead zone, the learnable region, and the mastered tail; published learnability curricula are its $N{=}2$ slice.

2. **A derived teacher and a characterized recycler** (§4–5): FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract is characterized as a verified, adaptively selected auxiliary update — on-policy for the achieved task only under an explicit distribution-matching condition (Remark 3).

3. **An estimator-conditioned coverage ordering, retired in its strong form and confirmed in its measured one** (§6): our first pre-registered factorial retired the cohort's zero-exception divergence at its registered endpoint; the time-integrated ordering that survived it was then registered as the primary of a second factorial on fresh seed blocks and confirmed 6/6 under both samplers (p=.031 each; 24/24 paired blocks across both waves), with controls closing the scheduler-mismatch and variance-normalization readings (§6.3b).

4. **A new failure mode and a motivated mitigation** (§6.8–6.9): exact-verifier recycling improves mean accuracy while losing coverage — recycling-induced sharpening, replicated at three seeds — and gating

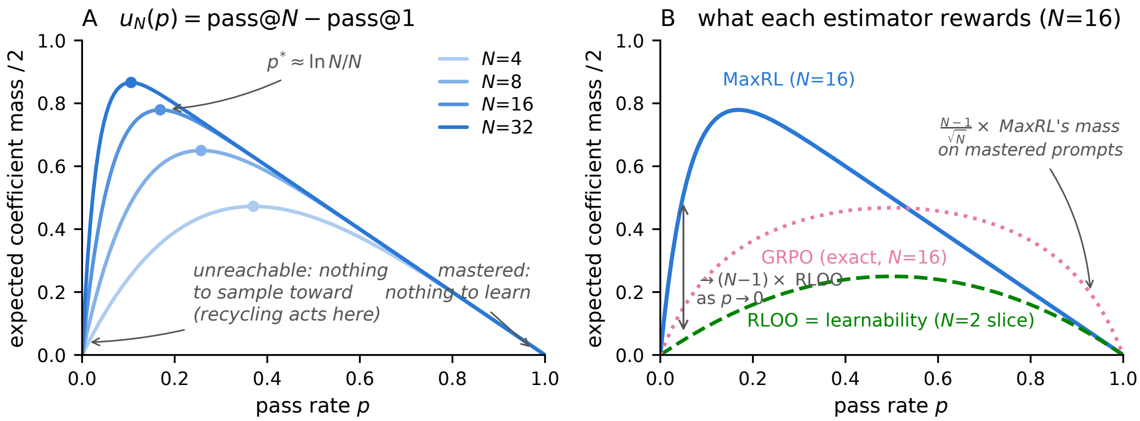

Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1-(1-p)^N) - p$ — half the estimator's expected coefficient mass $A_N = 2u_N$ (Prop. 1) — for growing rollout budgets N. The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; the utility vanishes exactly at $p \in \{0, 1\}$. *Right:* all three curves are exact expected half-mass at N=16 for the estimators *as deployed* (GRPO with sample-SD normalization: $\sqrt{\frac{N-1}{N}}\,\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$, $K \sim \mathrm{Bin}(N,p)$; MC-verified against the code). MaxRL concentrates $\to (N-1)\times$ more mass than RLOO on frontier prompts (Prop. 2); in the mastered tail MaxRL and RLOO decay together at first order (both $\to q = 1-p$; MC artifact `verify_guidance_math.json`) while GRPO's variance normalization keeps $\sim(N-1)/\sqrt{N}\times$ their relative mass there — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO's profile is exactly the published "learnability" objective — the N=2 slice.

relabels by an achieved-destination saturation statistic yields an operating point that restores frontier-tier coverage while keeping most of the mean gain.

Along the way we state a refinement to the MaxRL paper — the practical drop-K=0 estimator is unbiased for the T=N−1 objective, not T=N (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§8) and audited in the repository.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1-p)$ and truncates at order T, giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE (T=1) to exact ML ($T \to \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \to 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator is the curriculum

Lemma 1 (Truncation order of the practical estimator)**.** *Let $J_T(p) = -\sum_{t=1}^{T}(1-p)^t/t$ denote the T-truncated ML objective, so $\nabla J_T = \big(\sum_{j=0}^{T-1}(1-p)^j\big)\nabla p$; $T\to\infty$ recovers exact ML and T=1 REINFORCE. Under N i.i.d. binary rewards with pass rate p, the drop-K=0 variant of the practical MaxRL estimator has expectation $\big(\frac{1-(1-p)^N}{p} - (1-p)^{N-1}\big)\nabla p = \big(\sum_{j=0}^{N-2}(1-p)^j\big)\nabla p$: it is exactly unbiased for J_{N-1}, not J_N, for $N \geq 2$. Zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing the $-(1-p)^{N-1}\nabla p$ correction, and $\sum_{k=1}^{N}(1-p)^{k-1} - (1-p)^{N-1} = \sum_{k=1}^{N-1}(1-p)^{k-1}$.*

Interpretation. *Three estimators must be kept distinct: the* raw *success-conditioned average ($\frac{1}{K}\sum r_i S_i$, no control variate — the object of Tajwar et al. [1]'s Theorem 2, exactly unbiased for J_N), the* full control-variate *form (subtracting the unconditional mean score on every group, including all-fail ones), and the* practical drop-K=0 *form above, which* *is what ships. The lemma is a refinement to Tajwar et al. [1], immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped. Magnitude at the deployed budgets: the relative gradient shortfall of $T{=}N{-}1$ vs. $T{=}N$ is $(1-p)^{N-1}/\sum_{k=1}^{N}(1-p)^{k-1}$ — at $N{=}16$ it is 5.8% at $p{=}.01$, 1.1% at the band peak $p^*{\approx}.17$, and $<10^{-4}$ at $p{=}.5$; at $N{=}8$ it reaches 12% on the deep frontier. Bookkeeping at moderate p, material exactly where the frontier teacher concentrates sampling.*

Proposition 1 (Expected coefficient mass)**.** *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights, define the utility $u_N(p) := pass@N(p) - pass@1(p) = (1-(1-p)^N) - p$. Then the expected absolute coefficient mass is*

$$\boxed{A_N(p) \;:=\; \mathbb{E}\Big[\textstyle\sum_i |w_i|\Big] \;=\; 2\big(pass@N(p) - pass@1(p)\big) \;=\; 2\,u_N(p).} \tag{1}$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1-\frac{K}{N})$, and $\mathbb{E}[2(1-\frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(pass@N - pass@1)$. The sampling utility used everywhere below (figures, Algorithm 1, code) is u_N; the factor two matters only when quoting exact mass, and we carry it explicitly as $A_N = 2u_N$. Two exact companions tie mass to the gradient. Factorization*: writing $\mu_\pm(x) = \mathbb{E}[S \mid r{=}1/0, x]$ for the success/failure-conditioned score means, exchangeability gives, for every group size,*

$$\mathbb{E}[\hat{g} \mid x] \;=\; u_N(p)\,\big(\mu_+(x) - \mu_-(x)\big), \tag{2}$$

so u_N is exactly the estimator-controlled scalar multiplying the prompt's success–failure score contrast — the part of the expected update a sampler can know before rolling out; the contrast (and where it transfers) is the part it cannot. Weight view*: since $\mu_+ - \mu_- = \nabla p/(p(1-p))$, Eq. (2) is equivalent to $u_N(p) = p(1-p)\,w_{N-1}(p)$ with $w_T(p) = (1-(1-p)^T)/p$ the truncated weight function of Lemma 1 — mass and the practical estimator's weight view are linked, not identical objects. (Both identities MC-verified; artifact `verify_guidance_math.json`.)*

Interpretation. *The estimator's expected coefficient mass on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. Coefficient mass is a* surrogate *for learning signal (Remark 1), but its two zeros are exact, and they anchor three regimes of difficulty. A* dead zone *at $p = 0$: nothing to condition on; sampling alone cannot produce a success there.*[2] *A* learnable region *with a unique interior peak at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ (unique because $u_N'' < 0$). A* mastered tail *at $p = 1$: no contrast left. The regimes are continuous — the only exact zeros are $p \in \{0, 1\}$ — so where we quote a "band" we mean the threshold set $\{p : u_N(p) \geq \eta\, u_N(p^*)\}$ ($\eta{=}0.5$ unless stated), whose location and N-scaling follow from the one parameter you already chose. Everything in this paper is one of three moves against this structure: allocate within the learnable region (§4), create signal at its bottom (§5), and refuse updates that land in the mastered tail (the gate).*

Proposition 2 (Frontier amplification)**.** *As $p \to 0$, the ratio of MaxRL's expected mass to RLOO's ($A_N^{\mathrm{RLOO}} = 2p(1-p)$ exactly, under the $1/N$-normalized leave-one-out weights) tends to $N - 1$.*

Interpretation. *The comparison to GRPO is two-sided and exact* for the deployed implementation*, which normalizes by the* ***sample*** *standard deviation ($s_K = \sqrt{K(N{-}K)/(N(N{-}1))}$, PyTorch's default), not the population SD: $A_N^{\mathrm{GRPO}} = 2\sqrt{\frac{N-1}{N}}\,\frac{1}{N}\mathbb{E}\sqrt{K(N{-}K)}$ with $K \sim \mathrm{Bin}(N, p)$ (MC-verified against `estimators.py`). Taking tails, MaxRL/GRPO $\to \sqrt{N}$ as $p \to 0$ while GRPO/MaxRL $\to (N{-}1)/\sqrt{N}$ as $p \to 1$ (at $N{=}16$: 4.0× and 3.75×; the symmetric $\sqrt{N{-}1}$ holds only under population-SD normalization). MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sim(N{-}1)/\sqrt{N}\times$ more of its mass on mastered prompts — maintenance a curriculum then removes. One caveat governs every cross-estimator magnitude in this paper: each estimator's coefficients carry a conventional global scale (absorbable into the learning rate), so absolute ratios like these are facts about the implementations as shipped under a common learning rate, not invariant properties of the objectives. What survives any recalibration is the* shape*: where each estimator's mass concentrates as a function of p, and its exact zeros — which is why the empirical claims are about tails and signs, not magnitudes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.*

[2] At small $p > 0$ a task is *operationally* dead at budget (B, N) when its live-group probability $1 - (1 - \ell_N(p))^B$, with $\ell_N(p) = 1 - p^N - (1-p)^N$ and B its share of groups, is too small to sustain training. On the frontier-heavy pool of §6.2 ($p \leq 10^{-5}$, $\sim$133 groups per task) that probability is $\leq$2% per task, and the one live group observed across all uniform seeds did not ignite — dead in the budgeted sense, not by fiat.

The static known-p allocation problem this utility induces has a clean water-filling solution, but the deployed teacher does not use it (it keeps fixed group sizes and samples $\propto u^\gamma$ with a floor, trading myopic optimality for group-parallel batching and exploration); we state the allocator and its boundary conditions in App. B and measure what the gap costs in §6.1, where a no-floor, γ-matched true-pass-rate oracle ties the Thompson teacher in end performance.

Remark 1 (Scope of the surrogate). Coefficient mass counts weighted rollouts, not score-vector geometry. We use it as the sampling utility because (i) it is knowable *before* spending rollouts, (ii) its zeros are exact — at $p \in \{0, 1\}$ the estimator emits no gradient of any norm — and (iii) on the exact-gradient testbed it *ties* the true first-order expected improvement as a task ranking (Spearman within .005 at every snapshot, and exactly 1.0 against the full exact transfer matrix — one expected group per task, exact Δp on every other task — at three training snapshots; artifacts `results_bridge.json`, `results_transfer_matrix.json`). That tie is a property of *shared-prefix* task structure, where activity and transfer align; on pools without it, activity ranks what a task can emit, not where it helps (§6.5–6.6 measure that gap). What it does not claim to be is *the* optimal sequential utility, and no state-local functional is: on independent tasks the true final-performance optimum is a planning solution (trajectory-knapsack with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^\alpha u_N$ wins anytime performance while model-predictive replanning wins the final checkpoint, each losing the other's metric (8–10/10 paired seeds each way; App. C). "Best curriculum utility" is ill-posed until that objective is fixed; we report both currencies throughout. The claim that carries the section is the pair of exact zeros at $p \in \{0, 1\}$, which every candidate utility shares — though the zeros alone do not suffice: a uniform-over-band control (sample the in-band indicator, discard the shape) forfeits most of the allocation gain on chains (0.681 vs. 0.743 AUC, 0/10 seeds; artifact `results_bridge.json`). u_N's standing is therefore: the shape matters, the best shape is objective- and testbed-dependent, and u_N is the one shape that ties the exact first-order objective as a predictor, requires no tuning, and transfers to the maze within noise.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x\sim q_t}[J(x)]$: each group's gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ. This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t{=}\rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ, so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ-measured outcome.

One identity, three literatures: learnability curricula [11, 29] are the $N{=}2$ slice of u_N; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [15] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). DisCO [2] and SC-SDPO [3] derive analogous per-question weights for the GRPO family to diagnose difficulty bias; our identity generalizes that line to success-conditioned estimators and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The three regimes, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the learnable region — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal where sampling the requested task yields none — in our control battery, the one intervention that still adds on top of the oracle. The **estimator** underneath shapes whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe.*

4 FrontierMax: the schedule

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior tracks each prompt's pass rate from observed group outcomes only — never from relabeled successes, which inflate it ($\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a uniform floor. The knobs (decay 0.7, floor 0.1, $\gamma \in \{1, 4\}$) carry validated defaults (Table 3); what is *derived*

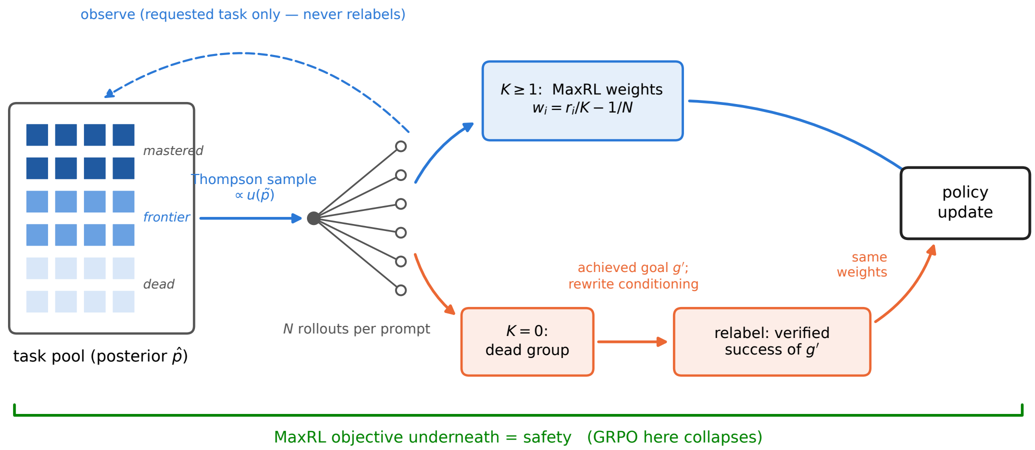

Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked ▷).

▷ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \mathrm{Beta}(\alpha_x, \beta_x)$ (else: uniform)
roll out N completions per prompt; verify; $K_x \leftarrow$ successes
update posterior (α_x, β_x) from (K_x, N) *for the requested task only*
live groups $(1 \leq K_x)$: MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
▷ **dead groups** (K_x=0): relabel each parseable failure to *its own* achieved sub-goal g' (verified; conditioning rewritten); unparseable rollouts stay failures, so the group retains contrast $(K < N)$
▷ admit the relabel only if $\hat{f}_{g'} \leq$ `gate_max_p`, where $\hat{f}_{g'}$ is a decayed frequency of g' in the recycler's recent relabel stream (recorded whether or not admitted) (the gate: refuse recently seen destinations; never-seen destinations start at the Beta(1,1) prior $\hat{f} = .5$, admitted at the default threshold)
one policy-gradient step on live + admitted relabeled groups

---

rather than tuned is the difficulty band itself — location, width, and N-scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Remark 3 (Hindsight update, characterized). Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields a *verified weighted-likelihood update for the achieved task*: every relabeled example is a true success of g' under the environment's verifier. It is generally an *adaptively selected, off-policy auxiliary update*, not an unbiased gradient of g''s on-policy objective: g' is selected from the same group being scored, conditioning on the requested task's failure correlates the rollouts, the conditioning rewrite changes the law under which the log-probability is evaluated, no importance ratio corrects for it, and the finite-N weights target J_{N-1} (Lemma 1) rather than exact ML. It equals the fresh-rollout J_{N-1} gradient for g' only under a distribution-matching condition — the selected, rewritten group must have the same joint law as an i.i.d. fresh rollout group for g'. This specializes known relabeling-bias structure — the induced-distribution view of Eysenbach et al. [20] and the conditional-law gap in Ghosh et al. [21] — to success-conditioned RLVR weights; we state it as a remark, not a result, and note that the sharpening phenomenon of §6.8 runs through the *marginal* shift those works also identify, not this conditional matching. One further gap separates this remark from the deployed LLM loop: the remark is written for a group with *one* destination g', but Algorithm 1 relabels each parseable

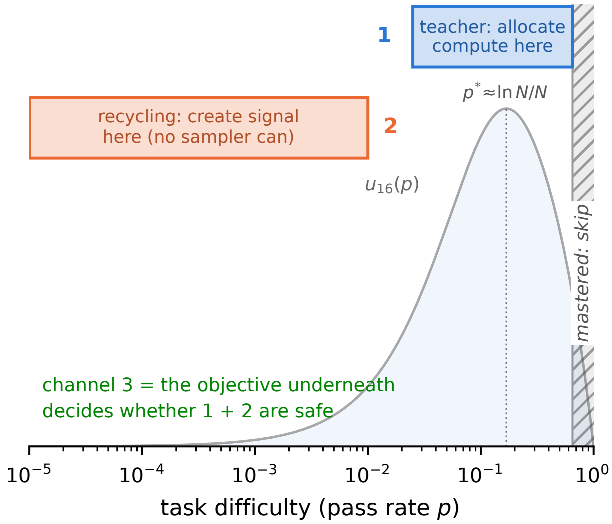

Figure 3: **The three regimes as a map of the method.** The teacher allocates compute within the learnable region; recycling creates signal where sampling the requested task yields none; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

failure to *its own* achieved goal, so one group can mix rewritten tasks — and once destinations differ within a group, the shared success count K couples unrelated tasks and the update is not a J_{N-1} gradient for *any* single achieved task. We therefore read the deployed per-row variant as a *verified weighted-SFT update*, treat its results as evidence about that objective, and measure the coupling's cost where all variants are exactly implementable: per-row relabels as their own K=1 groups lead (AUC .952), one-destination-per-group is next (.881), and the shared-K coupled form forfeits most of recycling's value (.749 vs. no-recycling .705; both orderings 10/10 paired seeds; artifact `results_row_vs_group.json`). Full object-level characterization, the CPU-reference contrast, and the LLM-side ablation flag: App. A.

Interpretation. *"Exactness" is a property to* measure per environment*, not a blanket guarantee. We measured the two gaps the remark names where they are measurable (exact chains, 5 seeds; artifact* `results_proposal_shift.json`*): the applied relabel update's cosine against the exact fresh-destination gradient is* 0.93*; the selection bias on the destination's success rate is small here (*+.003*); and cross-fitting the destination choice removes the adaptive reuse at a real cost (alignment* → 0.82*, AUC* .883 → .829*). The off-policy caveat is real; its sign is environment-specific.*

Two contracts keep practice on the proof's side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment's own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Among the interventions studied here, recycling is the one that *creates* signal for an all-fail requested task (adding the achieved sub-goals to the pool by hand would too — what recycling buys is discovering them, §6.2), and we decompose its headline gain rather than let it inflate: of the +0.22 AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries +0.037 beyond the strongest placebo, is the tightest arm in the battery, and is exactness-dependent (random-target relabels lose to dose-matched replay 5/5). Both numbers ship together. The value is regime-dependent in a way we can state precisely: +0.22 AUC on fixed task pools, +0.01 on one-shot task streams — the gain is proportional to how much a relabeled skill can compound, and task-set fixedness is the single most important regime variable we found.

Table 1: The ladder at a glance: every rung asks one question of the three-regime structure; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3a	does the curriculum pay?	maze cohort, 1.26M	3	throughput + recycling signal
6.3b	estimator effect, real grads?	maze factorial	6 blocks	endpoint failed; expl. AUC 12/12
6.4	where/when does coverage go?	maze, level-resolved	22 runs	easy band, immediately
6.5*	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6*	beyond task spread?	MountainCar/CartPole	3	spread=existence; tea.=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	sign landed; replication open
6.8	what does recycling cost?	Countdown v2	3	mean up, coverage down
6.9	does the gate recover it?	Countdown v2	3	one useful trade point

*boundary rungs, compressed to one paragraph in the main text; full protocols and figures in App. C.

6 Experiments: an escalating ladder

Each rung asks one question of the three-regime structure, at increasing scale and decreasing experimental control; Table 1 is the map. Statistical contract, stated once and used throughout: the independent unit for any training-method claim is an independently trained seed block; runs sharing a seed, warmstart, or checkpoint are paired or correlated observations, never additional replicates; levels, k values, metrics, and checkpoints from one trained model are repeated measurements of it; repeated-evaluation spread measures evaluation noise, not training-seed uncertainty, and every ± names its source at point of use (± with no qualifier is standard deviation across training seeds). "Coverage" is pass@k with exactly k samples drawn per prompt and averaged over the eval pool, at the k named per suite (8 on the maze, 16 on Countdown, 4 on GSM8K); mean@k is the mean success fraction of the same k samples; AUC is the mean of the eval curve over training at matched wall-clock or episode budget, and final is the last checkpoint. Where run pools are heterogeneous we say so and report signs descriptively rather than attaching p-values (§6.3–6.4). Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository's audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform 0.650 → teacher 0.728 → full stack $\mathbf{0.890 \pm 0.002}$ AUC. An eight-arm control battery (App. C) orders into a *gradient-information ladder*, each rung a 5/5 paired-seed win: zero-information dose (0.798–0.800) < replayed correct direction (0.832) < true relabels (0.869) < perfect allocation (oracle 0.8885 ≈ full stack 0.8895) < allocation+creation (0.8935). Two decompositions matter downstream. *Allocation saturates at the top of the ladder, not at the teacher*: a no-floor, γ-matched true-pass-rate oracle (0.8885±.0014) sits within the five-seed variation of the *full stack* (0.8895±.0023) — similar within observed noise, not tested for equivalence. The posterior-only teacher scores 0.728, far below the oracle, so perfect difficulty information is worth a great deal over the posterior alone; the accurate reading is that *recycling compensates for the posterior's allocation error* — the combination reaches what perfect information reaches, by a different route. *Creation decomposes*: a replay placebo captures 83% of recycling's +0.22; the relabel *direction* carries +0.037 beyond it and is the most reliable arm in the battery, while random-target relabels lose to dose-matched replay 5/5. The utility form matters through its N: the N-blind learnability slice $p(1-p)$ forfeits the entire allocation gain (−.007 vs. uniform) where u_N carries +.078 (5/5). All arms converge within .027 of one another — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling.

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; every intervention buys speed, not asymptote.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5}; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ∼400 groups, then goes silent. The attribution control pins down which channel did it: *uniform*+recycling

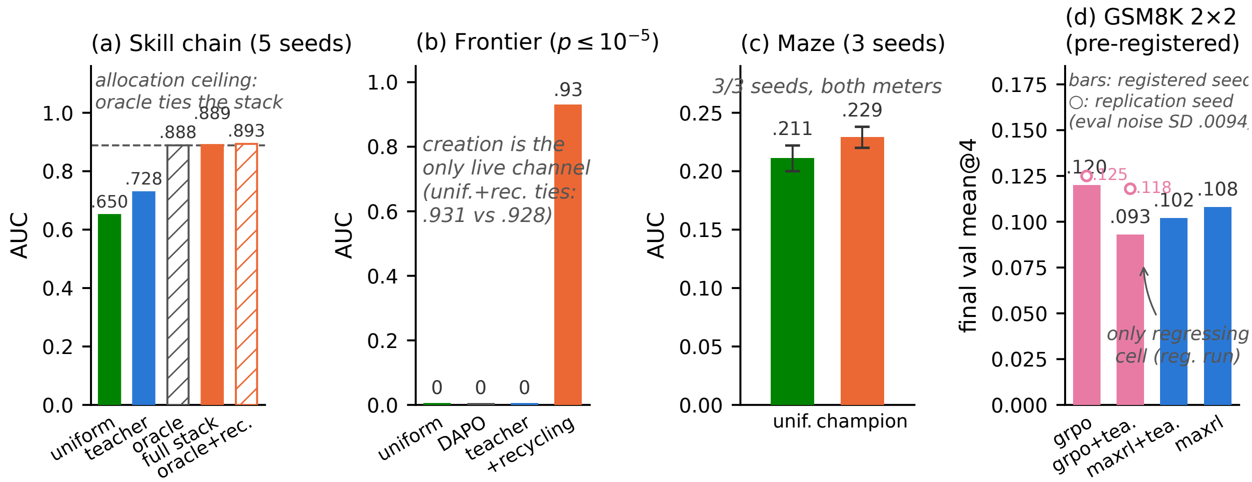

Figure 4: **The ladder.** (a) Exact-gradient chains: a no-floor, γ-matched true-pass-rate oracle matches the full stack within the observed five-seed variation — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 3/3 seeds on both meters. (d) GSM8K 2×2: bars are the registered run, open markers the replication seed (GRPO cells; eval-noise SD .0094 quoted in-panel): GRPO+teacher is the only regressing cell on the registered run (pre-registered P-G2; the replication seed did not reproduce the shape — §6.7); the teacher's MaxRL effect is a null at this budget (P-G1).

scores the same (0.931±.002 vs. teacher+recycling 0.928±.003 AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy's own failures, with no task engineering. One loss-level alternative remains: the MaxRL paper's own Eq. (10) keeps the control-variate term $-\frac{1}{N}\sum_i S_i$ on all-fail groups, so that estimator is *not* silent at $K{=}0$ ($\mathbb{E}[\hat{g} \mid K{=}0] = \nabla p/(1-p)$, MC-verified). We ran it (5 seeds, matched budget; artifact `results_fullcv_baseline.json`): on this pool the full-CV variant spends 100% of its update norm on all-fail groups and still scores 0.000 in every seed, with or without the teacher — the all-fail term pushes down every sampled action symmetrically and cannot ignite a pool this deep, while recycling under either variant reaches 0.98. On the balanced pool the two variants tie (paired ΔAUC straddles zero, 5 seeds).

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Even the estimator family's nonzero all-fail update (full control variate) stays at zero here: creation is the only channel we tested that ignites a dead pool, and its value is discovery, not allocation.

6.3a Does the curriculum pay on the maze? (transformer at matched wall-clock; 1.26M params, 13 levels, 3 seeds). Champion (teacher + dense recycling) 0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive in 3/3 seeds on both meters (six correlated contrasts, not six seeds); 22–35% more optimization steps per GPU-hour. Attribution: matched *wall-clock* lets curriculum arms take 1.2–1.4× more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher's effect (step-matched teacher-only ΔAUC +0.006, within seed noise) while teacher+recycling survives step-matching, positive in all three paired seed blocks[3] — on the maze the teacher's measured value is *throughput*, recycling's is signal. Utility-form

[3] A paired t at $n{=}3$ gives $p{\approx}.04$, reported as descriptive given its normality sensitivity; an exact sign-flip test cannot go below .25 at three blocks.

controls (3-seed reruns, artifact `un_form_verdicts.json`): the exact u_N measures within noise of the legacy deployed form on the maze itself, and a CPU-winning $(1-p)$-tilted refinement loses 0/3 here — within-band utility choices are testbed-sensitive, which is why the theory claims only the band's boundaries.

6.3b Does the estimator effect survive real gradients? (the pre-registered factorial). An earlier nine-run cohort showed perfect sign separation — every MaxRL-labeled run grew pass@8, every GRPO-labeled run lost it — but the pool mixed sampler configurations and hindsight doses and reused warmstarts, so we pre-registered the clean test: a balanced $\{$MaxRL, GRPO$\} \times \{$uniform, u_N-teacher$\}$ factorial, six independent seed blocks, identical per-block warmstarts, matched generation-and-optimization budgets (250 fixed steps), one registered endpoint (paired Δ mean pass@8, final minus warmstart), and a committed falsification branch. **The endpoint failed: 3/6 paired blocks under uniform, 4/6 under the teacher, and only 4/12 MaxRL cells grew coverage at all — the cohort's zero-exception divergence is retracted, not softened** (prereg + raw cells + verdict: `curriculum_maxrl/maze_gpu_factorial/`, vendored with execution-commit provenance). The cohort pattern had conflated recycling's coverage contribution with the estimator effect: its MaxRL runs included hindsight arms, and the factorial isolates the estimator.

What survives, at its evidence level. *Confirmed, exact rung*: frozen realized schedules replayed under each estimator preserve MaxRL over GRPO in 10/10 seed×schedule pairs (artifact `results_schedule_matched.json`) — the ordering is robust where gradients are exact. *Found exploratorily, then registered and confirmed on fresh randomness*: on time-integrated coverage (mean in-training pass@8 minus warmstart), MaxRL retains more coverage than GRPO in **12/12** wave-1 paired blocks — a noise-integrating secondary we had not registered, so we pre-registered it as the primary of a confirmation wave on six fresh seed blocks (P-F2, $\geq$5/6 under both samplers, committed falsification branch) and it **confirmed at 6/6 under each sampler** (exact two-sided sign p=0.031 per sampler; means +.015/+.024), with the easy-band concentration replicating at 10/12 pairs (P-F3; GRPO's easy-band loss $\sim$2× MaxRL's) — where the mass curves say GRPO overspends. Across both waves the ordering stands 24/24 paired blocks. The retired *endpoint* contrast, unregistered on the fresh blocks, landed 5/6 there — consistent with the power analysis (§8) and left retired; the claim this paper makes is the time-integrated one.

Three controls close alternative readings. *Scheduler mismatch*: GRPO driven by its *own* exact mass functional ($\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$, Thompson-sampled) does not close the gap on chains (5/5) and loses coverage like every other GRPO arm in the factorial (5/6 cells negative, mean −.042, equal to uniform-GRPO) — no scheduler choice rescues the estimator. *Variance normalization*: on chains, removing GRPO's SD normalization drops it to RLOO's coverage profile exactly as the mass account predicts (.148 vs. RLOO .161; sample-SD .762) — but in the factorial the no-SD arm loses the easy band in 6/6 seeds, indistinguishable from sample-SD GRPO, so **at neural scale the easy-band liquidation is not explained by variance normalization alone**: the Fig. 1 mechanism holds per-task under exact gradients and does not transfer through shared parameters unmodified (committed revision, falsification branch P-G0c). *Interaction*: the teacher does not amplify GRPO's coverage loss in the factorial (teacher−uniform Δcov@8 under GRPO: mean +.015, mildly protective) — the GSM8K P-G2 regression pattern receives no corroboration from the maze at this budget. Inference currency (single checkpoint pair; convention and per-level detail in App. C): our checkpoint needs **11**× fewer samples at level 5, ties GRPO at levels 2–3 — the summed-coverage curves cross at $k\approx4$, so choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO's coverage go? (level-resolved; cohort data — read with §6.3b's retraction). This section decomposes the *original heterogeneous cohort* by difficulty band; after the factorial, its value is the location and timing of GRPO's loss (which the factorial's easy-band asymmetry corroborates, 9/12 pairs), not the retired zero-exception contrast (Fig. 5). *Where:* GRPO's pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 (−.224 mean over its four runs, every run negative) while its L1–3 pass@1 *rises* (+.02 to +.14, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat (+.005±.058 over 18 runs) and concentrates its gain on the frontier band L4–6 (+.126±.063 vs. GRPO's +.057). The 18-run MaxRL pool is a heterogeneous run inventory (teacher variants, hindsight doses, a wide model, repeated seeds), so we report the band asymmetry descriptively — every

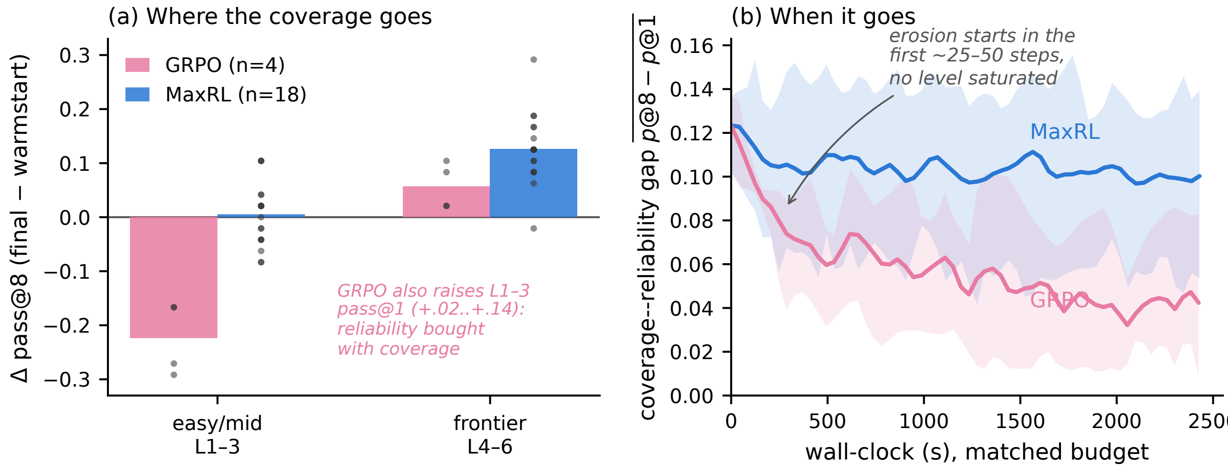

Figure 5: **Where and when GRPO's coverage goes (maze, level-resolved, matched wall-clock).** *(a)* Δpass@8 (final − warmstart) by difficulty band: GRPO's loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; a heterogeneous run inventory — descriptive, no p-value). *(b)* Coverage–reliability gap $\overline{\text{pass@8} - \text{pass@1}}$ vs. wall-clock (mean and min–max across runs): GRPO's erosion begins within the first ∼25–50 steps, before any level saturates; MaxRL stabilizes after warmup.

GRPO run negative on L1–3, the MaxRL spread centered at zero there — and attach no permutation p-value; the 22 runs are not exchangeable replicates of one estimator contrast. This is consistent with the mass curves in Fig. 1 — the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts where u_N has already released them — but the step from mass to coverage is a stated *hypothesis*, not a derivation: mass is sign-blind, and "mastered-tail updates sharpen" requires an argument about the sign structure of near-saturated groups that we support with the location and timing evidence here rather than prove. Its committed falsifier (remove variance normalization and the tail signature should vanish) split across the rungs, as §6.3b reports: the mechanism account is per-task-exact only, and we bound the claim accordingly. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total coverage–reliability gap loss ($\overline{\text{pass@8} - \text{pass@1}}$: .123 → .042) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only .62–.75; MaxRL's premium stabilizes after warmup (.122 → .101). A dose-response note from the same logs (descriptive, same caveat as above): sparse relabeling (1/dead group) uniquely preserves the coverage–reliability gap ($\overline{\text{pass@8} - \text{pass@1}}$) relative to none and dense, while dense banks the same coverage into the best final pass@1 — dose sets *where* the coverage dividend is spent, not whether coverage is lost.

Takeaway. The easy band is where the estimators differ — GRPO liquidates it for reliability from the first updates, and the factorial's exploratory read keeps that asymmetry (GRPO's easy-band loss 2× MaxRL's, 9/12 pairs) — but the mechanism attribution to variance normalization holds only at the exact rung. Coverage is the currency that sees all of it.

6.5–6.6 Two boundary rungs (compressed; full protocols, figures, and attribution in App. C). An IsaacLab Anymal-C pilot (5 arms, one seed, independent team, pre-registered null) marks where allocation does *not* pay: on a terrain grid learnable everywhere at budget, the stock greedy curriculum beats the frontier teacher outright (.372 vs. .278) — with the maze step-matched analysis and Countdown, the regime pattern is **allocation pays where unlearnable-at-budget regions exist to avoid**. A Gymnasium control suite (binary task success, 3 seeds) marks the opposite boundary: training on the target task alone is a provable zero-update freeze (every group all-fail, so every group-relative estimator emits exactly zero), while the gated full stack reaches 1.000±0.000 on both official tasks (Fig. 9, appendix); attribution shows task spread supplies

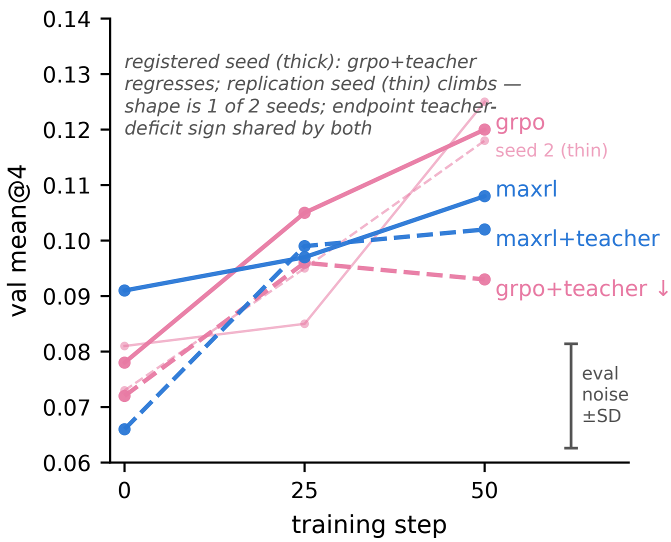

Figure 6: **GSM8K 2×2, val mean@4 over training.** Both GRPO seeds shown as paired trajectories (registered seed thick, replication thin); gray bar: repeated-eval noise floor of one fixed checkpoint (±SD .0094), drawn once as a scale. On the registered run GRPO+teacher is the only cell that regresses in its second half; the replication seed climbs instead. §6.7 gives the statistics and the steering telemetry that separates the two seeds.

gradient existence, the teacher speed and stability, and a per-bin-parameters ablation reproduces the known failure — **curricula operate through shared parameters, or not at all**.

6.7 Does it transfer to LLM scale? (GSM8K 2×2, pre-registered; SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half; the deciding statistic is the endpoint deficit against plain GRPO, 2–4× the measured same-model eval noise (mean@4 −.027; pass@4 −.048), while plain GRPO climbed monotonically (.078 → .105 → .120).[4] Steering telemetry, read against its own controls: the teacher's sampled-dead fraction *minimum* reaches .48 vs. uniform's .52, but the run means are indistinguishable (.66–.67 in all cells) — the delivered treatment was weak, which is the posterior-starvation mechanism below, and it bounds how much of the GRPO gap the curriculum can own. **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically (.073 → .095 → .118) under measurably weaker steering (near-uniform teacher on average). The completed within-seed pair (grpo-uniform seed 2: .081 → .085 → .125; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both (coupled) meters, with magnitude tracking steering intensity — a dose effect, not a sign flip; the pre-registered second-half-regression *shape* remains 1 of 2 seeds, suggestive not established. MaxRL+teacher trains stably to its best value (.066 → .102) but does not beat uniform's .108 — P-G1 is a *null at this budget*, predicted in advance by posterior starvation: the teacher's posterior learns this model's real difficulty landscape from ∼1 visit per prompt, but 3,200 draws over 7,473 prompts leave the estimated utility nearly flat, so Thompson sampling stays near-uniform (diagnostics in App. C). A pass@k sweep on the registered checkpoints has the teacher's GRPO deficit *widening monotonically with* k (−.008 at k=1 to −.024 at k=16) — curriculum damage lives in coverage, as the maze predicted, though each delta is within single-seed noise. One design confound is disclosed rather than dismissed: teacher cells sample with replacement and so revisit prompts more; a uniform-with-replacement control cell separates the repetition reading and runs with the steering-controlled replication. Small model; the pre-registered outcomes were orderings and signs on the registered run.

[4] The second-half dip itself (.096 → .093) is within the ±.009 noise floor and carries no weight on its own. One caveat governs every GSM8K number: the ±.009/±.017 bars are repeated-eval noise of a fixed checkpoint — they bound evaluation variability, not training-seed variability — so deficit-to-noise ratios are descriptive reads, not hypothesis tests.

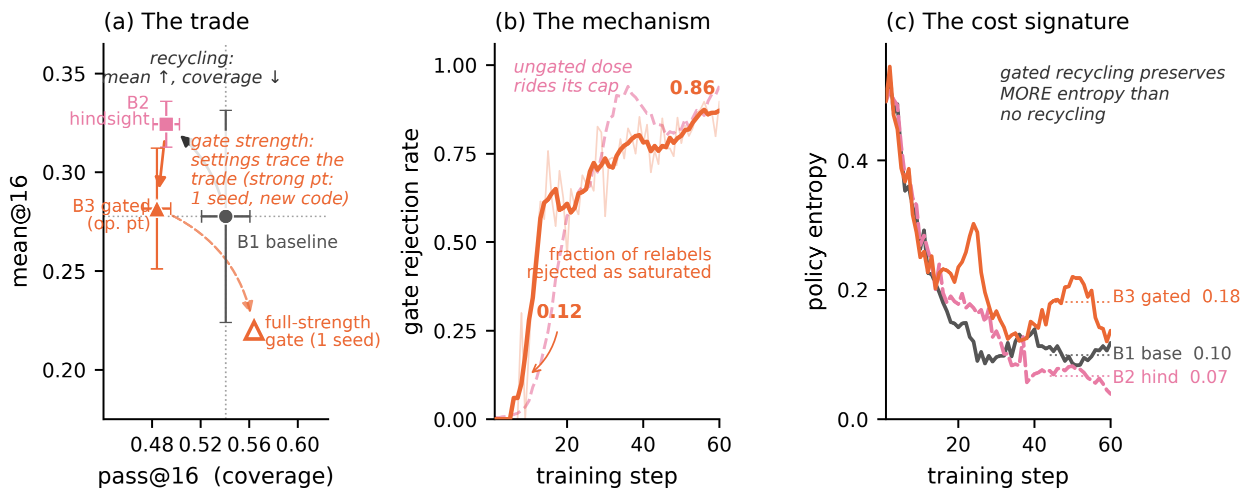

Figure 7: **Recycling-induced sharpening and its saturation-gate mitigation (Countdown, tier 1).** *(a)* Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage; the moderate gate (B3) gives most of the mean back on this early-saturating tier, and full-strength gating (hollow; 1 seed, corrected decay) returns coverage above baseline at the mean's full price — §6.9 states what these settings do and do not establish. *(b)* The mechanism in-run: the gate's rejection rate rises monotonically as the model's reachable set saturates its relabel destinations, while the ungated dose rides its cap. *(c)* Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

Takeaway. The LLM-scale interaction is 1-of-2-seeds, tracks steering intensity, and is not corroborated by the maze factorial's interaction read — the decisive experiment is a steering-controlled multi-seed cell.

6.8 What does recycling cost? (Countdown $2{\times}2$, pre-registered): two nulls and a new phenomenon. Our first exact-verifier hindsight experiment in an RLVR loop ran in production (12 relabeled groups/step, 72–89% yield; exact-verifier relabeling itself has precedent outside RLVR — CodeIt, proof-tree relabeling — see Related work). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool's simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §6.3b's weight-induced collapse — data-induced rather than weight-induced. The underlying concentration pathology is known in goal-conditioned RL (Skew-Fit's motivating failure mode; GCSL's marginal shift); what is unreported, to our knowledge, is its measurement in RLVR — no prior work quantifies the pass@k cost of hindsight relabeling in a verifier loop. Our algebra flags the mechanism: the achieved-goal marginal overweights destinations the policy already reaches, so relabel updates concentrate toward the mastered tail, where $u(p)$ says fresh training signal vanishes — the softmax pays for those updates out of the exploration tail.

Takeaway. Recycling needs an admission rule, and the utility's mastered-tail zero motivates one — designed and tested next.

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). *The gate, and what it is not.* The rule we implement is a saturation heuristic: a decayed frequency statistic $\hat{f}_{g'}$ over the recycler's own recent relabel destinations, thresholded at `gate_max_p`. It is *not* an estimate of the destination task's fresh-rollout pass rate $p_\theta(\text{success} \mid g')$ — the gate never rolls out g' and never observes a g' failure — and the

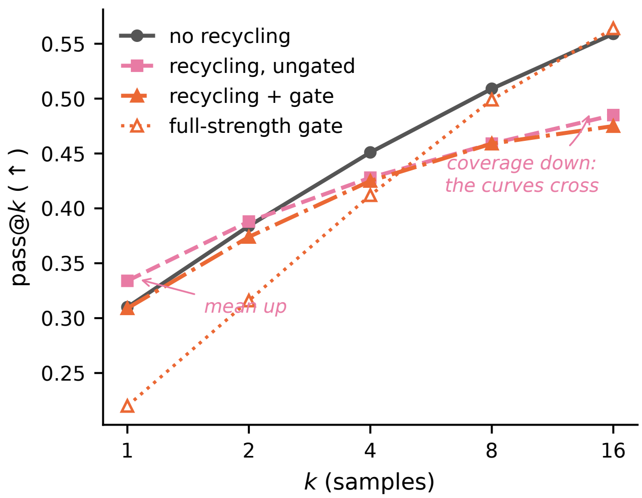

Figure 8: **Sharpening is a crossing in k (Countdown tier 1, step-60 val, seed 1; unbiased pass@k from 16 samples, n=128).** Ungated recycling sits above baseline at k=1 (mean up) and below by k=4 (coverage down), the deficit widening with k. The two gate settings bracket the trade — moderate tracks baseline at low k with a smaller high-k deficit; full-strength (hollow; one seed, corrected decay) pays at k=1 and finishes above baseline at k=16 — consistent with a strength dial, though §6.9 details why these settings do not yet establish one. Endpoints cross-check the 3-seed scoreboard; per-k telemetry from the seed-1 logs.

threshold 0.5 is a tuned midpoint, not a derived zero.[5] Where destination pass rates are exactly measurable (skill chains, 10 paired seeds; artifact `results_gate_variants.json`), the true-p gate preserves essentially all of ungated recycling's value (AUC .879 vs. .881) while the frequency heuristic pays a toll (.798; 10/10); at decision time the heuristic's statistic *anti-correlates* with true $p(g')$ (−.27 — it tracks the recycler's own recent output), and the two rules agree only once everything saturates (.66 → .99). The results below are therefore evidence about a recency-novelty filter that approximates saturation late in training; a genuine destination pass-rate estimate in the LLM loop is the indicated upgrade.

The experiment. We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms × three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). One setup note kept for its theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures, so there is nothing to recycle until SFT installs the output format (pool construction and baseline calibration in App. C). **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 → .324±.012) and pays −.049 of pass@16 (.541±.020 → .492±.011). The claim quantifies over k, so Fig. 8 shows it over k: the ungated arm starts *above* baseline at k=1 and crosses below between k=2 and k=4, with the deficit widening to k=16 — the crossing is the phenomenon.

The coverage loss clears both arms' seed spreads (> 2×); the mean gain clears the treated arm's tight spread but sits within the noisier baseline's — we report it as directionally consistent across seeds rather than individually significant. The frontier tier moves the same direction. **The timing is dose-then-damage:** recycling's frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ∼28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all seeds negative*, after ∼65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed; the gate's entropy retention replicates across seeds while ungated recycling makes entropy seed-unstable (details in App. C). **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline (.279±.019 vs. .274±.015) while keeping ∼60%

[5] u_N's exact high-p zero is at p=1, which would admit everything; a gate on a genuine task-conditioned $\hat{p}_{g'}$ is the derived object.

of the mean gain; at tier 1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely. Implementation disclosure: a decay-math bug made all three-seed gated runs gate *weaker* than designed (effective destination threshold $\hat{p} \approx 0.64$ where the design implies 0.92), so the reported operating point is an under-gated one; the corrected-decay rerun (1 seed) gates $\sim 3\times$ harder and pushes tier-1 coverage *above* baseline at the mean's full price (Fig. 7a). The three settings are *consistent with* a mean-vs-coverage trade — no gate: full mean gain, coverage lost; under-gated: $\sim$60% of the gain, frontier coverage restored; full-strength: gain erased, coverage above baseline — but they do not establish a monotone dose–response: the strong point is one seed, gate setting is confounded with the decay-code version, and at tier 1 the moderate gate moved *both* point estimates below the ungated arm. We claim the moderate frontier-tier setting as one useful operating point and defer the dial claim to the pre-registered fixed-decay sweep. The first designed-operating-point seed from that sweep (rejection fraction .93; artifact `armA_b3fix_s1.json`) lands at the recycling-off end — tier-1 mean *below* the no-recycling baseline — consistent with "strong gating $\approx$ no recycling" rather than a wide usable middle; remaining seeds are running, and if they agree, the usable range of the trade is narrower than the design intended. In-run telemetry shows the mechanism (Fig. 7b–c): rejection rises .12 $\to$ .85 as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the saturation gate buys back frontier-tier coverage at a measured mean price — one validated operating point, with the monotone-dial question deferred to a fixed-code sweep.

7 Related work

The coverage tension. RLVR itself trades coverage for precision: pass@1 rises while pass@k at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator's implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator. The estimator–difficulty interaction itself is a heating 2026 theme — SC-SDPO [3] shows GRPO's normalization implicitly encodes a difficulty weighting, and A-GRAE [28] redesigns the advantage around difficulty focus — but no prior work runs *identical* curricula under two estimators or measures the resulting pass@k sign flip.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1-\hat{p})$, which Prop. 1's corollary identifies as the $N{=}2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and the static water-filling reference (App. B) is the idealized form of both. DAPO's dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward pass@k but keep uniform sampling. **Goal selection in HER.** Selecting *which* goals to train on by difficulty or reachability is a decade-old idea in the relabeling literature: intermediate-difficulty goal generation [16], value-based hindsight goal selection (HGG; 17), curriculum-guided HER's proximity–diversity admission of relabel goals [18], and Skew-Fit's α as a concentration-vs-coverage dial [19]. Our gate belongs to this mechanism class; what is new is not admission-by-difficulty but its *motivation* — the same estimator functional u_N that schedules sampling says mastered-tail updates carry no fresh signal, so the teacher and the gate answer to one object — and the coverage accounting it is validated against. The gate statistic itself is a tuned heuristic (a decayed achieved-destination frequency with a tuned threshold, §6.8–6.9), so we claim a shared motivation, not a switch-free derivation.

Hindsight and self-training for LLMs. Exact-verifier relabeling has precedent outside RLVR loops — CodeIt [24] (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling [25] — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH [23] (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL's hindsight response replay [26] (densifying GRPO signal for multi-turn agents). None of these measures pass@k under relabeling — precisely the currency where we find recycling's hidden cost. The nearest phenomenon report is in same-problem rejection finetuning: Strozzi [27] document a STaR-trained model winning pass@8 while the base model overtakes at pass@64 — concentration over reach — for SFT on a model's own verified outputs. Recycling-induced sharpening is the cross-task, on-policy-RL sibling, and our gate is its estimator-motivated admission rule; gating *vanilla* RLVR updates by saturation signals has independent precedent (5's Bayesian boundary gating), not applied to relabel destinations. So the precise claims we retain, at the level our evidence and literature search support (we have not exhaustively searched concurrent 2026 work): to our knowledge no prior work (i) derives the sampling utility from the estimator's own expected coefficient mass, (ii) couples it with success-conditioned weights it provably matches, or (iii) *measures the pass@k coverage cost of hindsight relabeling in an RLVR loop* and tests a saturation-based relabel admission heuristic against that cost.

8 Limitations and negative results

Scale, and what the retraction taught about design. The factorial retraction is reported in full in §6.3b; what belongs here is its power lesson. Computed from all 36 runs' successive-eval differences, per-eval noise is $\sigma \approx .017$, giving the registered paired single-eval endpoint a contrast sd of .024 and power only 0.4–0.8 at the effect size the data itself displays (+.025/+.031 paired means). The time-integrated version of the same contrast cuts that sd by $\sqrt{10}$ (power ≈1 at +.02), which is why it became the registered primary of the confirmation wave on six fresh seed blocks — where it confirmed (P-F2, 6/6 per sampler; §6.3b). The LLM-scale teacher×estimator interaction is 1-of-2 seeds and tracks steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7); the maze factorial's interaction read (teacher mildly protective under GRPO) does not corroborate it. The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. A steering-controlled multi-seed LLM cell remains the decisive open experiment.

Where the teacher cannot work as specified. Two structural limits at production scale, stated as design constraints rather than future work. *Posterior starvation is the generic regime*: a per-prompt Beta posterior needs multiple visits, but realistic RLVR pools see ∼1 epoch (≤2–3 visits per prompt), so FrontierMax as specified is a small-pool instrument; deploying it on 10^5-prompt pools requires coarser state — bucket-level posteriors, a learned difficulty predictor, or warm starts — and our external annotations transferred poorly ($\rho = -0.17$), so cold-start from off-model labels is not the answer. *The throughput mechanism does not transfer*: at matched wall-clock the maze teacher's measured value is extra optimization steps from skipping dead groups' backward passes, but in generation-dominated LLM pipelines (85% of step time) a dead group has paid its full cost before it is known dead — the relevant LLM-scale comparisons are generation-budget-matched (resample-to-fill, prompt pre-filtering), and the teacher's value there rests on avoiding the generation itself, which requires the posterior it starves for. These two constraints compound, and we state them together because they bound the same deployment.

Scope of the surrogate. Coefficient mass is not gradient norm, SNR, or expected improvement (Remark 1) — it omits score-vector geometry, sequence length, token masking, optimizer state, and clipping — and §6.5 exhibits the regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment's relabel map; noisier verifiers will land between our fixed-pool and one-shot endpoints (+0.22 vs. +0.01 AUC).

The coverage main effect is pool-conditional. A pre-registered fifth suite (Jugs water-measuring puzzles; preregistration, per-cell raw endpoints, verdicts, and postmortem vendored under `jugs_llm/results/` with execution-commit provenance) returned all-null: on a pool whose learnable band is a single stratum of 1–2-move template solutions, *plain MaxRL* also collapses — entropy 1.36 → 0.07–0.21 in 60 steps, frontier-tier pass@16 0.15 → 0, every arm converging to the same deterministic policy — and recycling accelerates the collapse rather than rescuing it. When N rollouts of a near-trivial plan are near-identical, the i.i.d.-rollout diversity the group contrast requires is absent, and success-conditioned weights liquidate frontier coverage through shared parameters just as GRPO's do elsewhere. The estimator main effect of §6.3–6.4 is therefore a claim about pools with a graded band at the deployed N, not an unconditional safety guarantee; per-task rollout-set diversity at initialization is the design gate we now check first. The same suite also exposed a granularity bug in the gate (keying destinations by value collides tasks when achieved values are few and shared — ≤ 20 integers in Jugs vs. hundreds in Countdown); the corrected gate keys on the relabeled task.

What did not transfer. A 4× duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ-concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T; and feeding relabeled successes to the teacher's posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a "beats the oracle" comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected match-within-noise). Full postmortems live in the repository's audit trail.

9 Conclusion

For the practical success-conditioned estimator under i.i.d. binary rewards, the expected absolute coefficient mass of a prompt is exactly $2(\text{pass@}N - \text{pass@}1)$, and the drop-all-fail implementation targets truncation order $N-1$. That identity supplies a compute-aware sampling heuristic and motivates a saturation-based admission rule for verified relabels; it does not by itself prove a learning band, optimal prompt allocation, or on-policy hindsight, and we have been explicit about where each stronger reading fails — including against ourselves: the balanced estimator-by-sampler factorial we pre-registered retired our own strongest neural claim at its registered endpoint, this paper reports that retraction in place rather than re-grading it, and the ordering that survived was then registered on fresh seed blocks and confirmed (§6.3b). What the record supports: an exact-rung estimator-conditioned coverage ordering that survives frozen schedules and its own controls, its time-integrated neural-scale form — exploratory in wave 1 (12/12), registered and confirmed in wave 2 (6/6 per sampler, p=.031 each; 24/24 across both waves) — a three-seed replication of recycling-induced sharpening on Countdown with one gated operating point that buys the coverage back, and a pre-registered negative (Jugs) marking where the whole family of interventions fails together. The remaining confirmatory experiments — a fixed-code gate-strength sweep and a steering-controlled multi-seed LLM cell — are pre-registered in the artifact with committed falsification branches, and the factorial shows we execute those branches when they fire. The portable practice we advocate is independent of our method: measure coverage next to mean accuracy whenever a curriculum or recycler ships, conditioned on the estimator it ships on.

References

[1] Tajwar, F., Zeng, G., et al. Maximum Likelihood Reinforcement Learning. *ICML*, 2026.

[2] Li, G., Lin, M., Galanti, T., Tu, Z., Yang, T. DisCO: reinforcing large reasoning models with discriminative constrained optimization. *NeurIPS*, 2025.

[3] Liu, Z., Cao, Y., Chen, J., Honavar, V. G. Restoring the sweet spot: pass-rate weighted self-distillation for LLM reasoning. arXiv:2605.27765, 2026.

[4] Yue, Y., et al. Does RL really incentivize reasoning capacity in LLMs beyond the base model? *NeurIPS* (oral), 2025.

[5] Yuan, S., et al. Understanding diversity collapse in RLVR via the lens of overtraining. arXiv:2606.15455, 2026.

[6] Cui, G., et al. The entropy mechanism of RL for reasoning language models. arXiv:2505.22617, 2025.

[7] Cai, W., et al. Curriculum RL can incentivize reasoning capacity beyond the base model. arXiv:2606.22317, 2026.

[8] Wang, Z., Cui, G., Li, Y.-J., Wan, K., Zhao, W. DUMP: automated distribution-level curriculum learning for RL-based LLM post-training. arXiv:2504.09710, 2025.

[9] Shi, T., Wu, Y., Song, L., Zhou, T., Zhao, J. Efficient reinforcement finetuning via adaptive curriculum learning. *TMLR*, 2026. arXiv:2504.05520.

[10] Chen, X., et al. Self-evolving curriculum for LLM reasoning. arXiv:2505.14970, 2025.

[11] Foster, T., Sims, A., Forkel, J., Foerster, J. LILO: learning to reason at the frontier of learnability. *NeurIPS*, 2025.

[12] Yu, Q., et al. DAPO: an open-source LLM reinforcement learning system at scale. *NeurIPS*, 2025.

[13] Xiong, W., et al. Reinforce-Ada: an adaptive sampling framework under non-linear RL objectives. arXiv:2510.04996, 2025.

[14] Zheng, H., et al. Act only when it pays: efficient reinforcement learning for LLM reasoning via selective rollouts (GRESO). *NeurIPS*, 2025.

[15] Andrychowicz, M., et al. Hindsight experience replay. *NeurIPS*, 2017.

[16] Florensa, C., Held, D., Geng, X., Abbeel, P. Automatic goal generation for reinforcement learning agents. *ICML*, 2018.

[17] Ren, Z., Dong, K., Zhou, Y., Liu, Q., Peng, J. Exploration via hindsight goal generation. *NeurIPS*, 2019.

[18] Fang, M., Zhou, T., Du, Y., Han, L., Zhang, Z. Curriculum-guided hindsight experience replay. *NeurIPS*, 2019.

[19] Pong, V. H., Dalal, M., Lin, S., Nair, A., Bahl, S., Levine, S. Skew-Fit: state-covering self-supervised reinforcement learning. *ICML*, 2020.

[20] Eysenbach, B., Geng, X., Levine, S., Salakhutdinov, R. Rewriting history with inverse RL: hindsight inference for policy improvement. *NeurIPS*, 2020.

[21] Ghosh, D., et al. Learning to reach goals via iterated supervised learning. *ICLR*, 2021.

[22] Ferbach, D., et al. Self-consuming generative models with curated data provably optimize human preferences. *NeurIPS*, 2024.

[23] Xu, I., Jiang, S., et al. Learning more from less: reinforcement learning from hindsight. arXiv:2607.09042, 2026.

[24] Butt, N., et al. CodeIt: self-improving language models with prioritized hindsight replay. *ICML*, 2024.

[25] Poesia, G., et al. Learning formal mathematics from intrinsic motivation. *NeurIPS*, 2024.

[26] Hu, Z., Wang, L., Wang, X., et al. ZipRL: adaptive multi-turn context compression with hindsight response replay. arXiv:2605.28069, 2026.

[27] Strozzi, I. L. Teacher-free self-training amplifies but does not compound: a pass@K crossover on a free-verifier domain. arXiv:2606.07856, 2026.

[28] Yu, Z., Chen, Z., Liu, M., et al. Unveiling implicit advantage symmetry: why GRPO struggles with exploration and difficulty adaptation. arXiv:2602.05548, 2026.

[29] Rutherford, A., et al. No regrets: investigating and improving regret approximations for curriculum discovery. *NeurIPS*, 2024.

A Mixed-target relabel groups: full characterization

The deployed per-row LLM variant trains each relabeled row as a true success of its own rewritten prompt at weight $1/K - 1/N$, with two consequences we accept and disclose: the $-1/N$ failure weight acts as a conditional push-down on unparseable rollouts rather than a zero-mean baseline, and success weights couple across unrelated destinations. The CPU reference implementation instead trains each relabeled trajectory as its own K=1 group at weight $1 - 1/N$, which *is* the single-destination object of Remark 3; the per-row LLM variant was pre-registered as a deviation because Countdown groups rarely share achieved values. The single-destination MaxRL form — one destination per dead group, all N rows rewritten and reverified against it — is the design the remark licenses. Whether the deployed loop pays the chain-rung coupling penalty is a question that rung cannot answer (verl's normalization and failure conditioning differ); the artifact ships a single-destination ablation flag (`one_target_per_group`, CPU-tested) and its pre-registered LLM run for exactly that test.

B The static allocation reference

Proposition 3 (Allocation, static). *Consider the static problem* $\max_{\{N_i\}} \sum_i u_{N_i}(p_i)$ *subject to* $\sum_i N_i = B$ *and integer* $N_i \geq 1$ *(every prompt receives a mandatory initial rollout; feasibility requires* $B \geq$ *the number of prompts), with pass rates* p_i *fixed and known and one group per prompt. Because the marginal gain* $u_{N+1}(p) - u_N(p) = p(1-p)^N$ *is decreasing in* N*, greedy water-filling of the* $B - \#$*prompts remaining rollouts on this marginal — the probability that the next rollout is a group's first success — is optimal. The restriction to* $N_i \geq 1$ *is not cosmetic: the formula* u_N *extends to* N=0 *as* $u_0(p) = -p$*, but the true mass of an unallocated prompt is* 0 *(as is a single-rollout group's,* $u_1 \equiv 0$*), so with optional activation (*$N_i \geq 0$*) the objective is not* $\sum_i u_{N_i}(p_i)$ *and the problem acquires an activation structure outside this proposition's scope.*

Interpretation. *A mass-optimal* reference *allocator for the static, known-p problem only; it does not prove that sampling prompts* $\propto u^\gamma$ *is optimal (at fixed group size, expected immediate mass is linear in the sampling distribution, so the unconstrained myopic optimum is a brittle arg-max), nor does it describe the §6.1 oracle, which samples prompts by true-p utility at fixed group size rather than varying* N_i*. It is also the idealized common form of across-prompt reallocation (our teacher) and within-prompt rollout adaptation (Reinforce-Ada): water-filling spends its next rollout wherever* $p(1-p)^N$ *is largest.*

C Full details for compressed experiment rungs

Material moved from §6 during the page-budget pass; every number here is artifact-backed in the repository.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr×2 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ-matched oracle 0.8885±.0014, full stack 0.8895±.0023, oracle+recycling 0.8935±.0012. The relabel direction term (+.037 over replay) has seed spread ±.0025, 4–6× tighter than every placebo; its worst seed beats every placebo's best. Random-target relabels are harmful relative to dose-matched replay (−.032, 5/5 seeds) but neutral vs. plain lr-doubling: fake labels waste the slot rather than poison training. Utility-form controls: exact u_N 0.728±.002 vs. legacy frontier form 0.733±.013 (the derived form's payoff is 7× smaller seed variance); the N=2 slice $p(1-p)$ is a wash against uniform (−.007, 2/5 wins) where u_N carries +.078 (5/5). AUC spread across arms is 9× the final-performance spread.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated; artifact `maze_gpu/efficiency.json`; Fig. 10 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k\approx4$; 5.12 vs. 4.62 at k=64, with GRPO's level-2/4 curves saturating at 0.88/0.56 where MaxRL's reach 1.00/0.62.

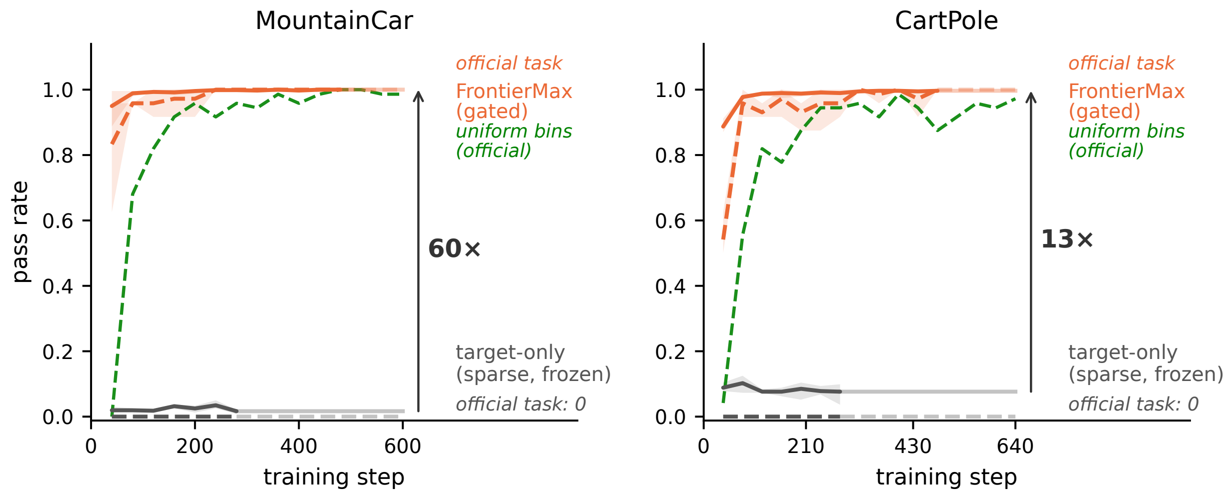

Figure 9: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward; a uniform-over-bins control (green) also cracks the task, 3× slower and off-ceiling; the gated FrontierMax stack converges to 1.000±0.000.

6.5 IsaacLab protocol. Five arms: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed, pre-registered protocol. The teacher's only edge over the greedy walker is a within-noise sliver at the hardest evaluated level.

6.6 gym control (full). MountainCar flag: measured random-policy success $< 3\times10^{-6}$ over 10^6 episodes; survive-400 $\approx 10^{-13}$ by tail fit. Dense-reward controls (prediction registered before the run): CartPole's native +1/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.96±0.03, 3 seeds); MountainCar's native −1/step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 → +recycling .895 → +teacher .956; CartPole .708 → .792 → .893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect 3.0× to hard-task 0.9 (2.1× to 0.99 on CartPole). The per-bin ablation replicates the 10-seed transition-matched study (flag 0.848±0.058) before strengthening it to full convergence.

6.7 posterior diagnostics and harness caveats (full). The pass@k sweep's absolute level sits ∼3× below the trainer's val metric (unreconciled harness discrepancy; only the within-sweep contrast is quoted). Posterior detail: two independent teacher runs' posteriors agree with each other (cross-run Spearman ≈.45 on co-visited prompts, $n\approx900$; ≈ .67 self-consistency with a shared latent) vs. .10–.22 against external 7B-model annotations, from ∼1.25 visits per prompt. Across external-difficulty quintiles spanning solve rates .33 → .98, mean $\hat{p}$ moves only .076 → .103 — the compressed landscape that starves allocation. A response-length read we advanced after five runs (three GRPO runs compressed harder than both MaxRL runs) did *not* survive the sixth: the within-seed control run finished longer than every earlier run (252 vs. 208–236 tokens from a common ∼365-token warmstart), so output length does not separate the estimators at this scale and we withdraw it as a sharpening signature. Prompt-uniqueness figures for the replacement confound: ∼34% unique prompts seen by teacher arms vs. ∼43% uniform.

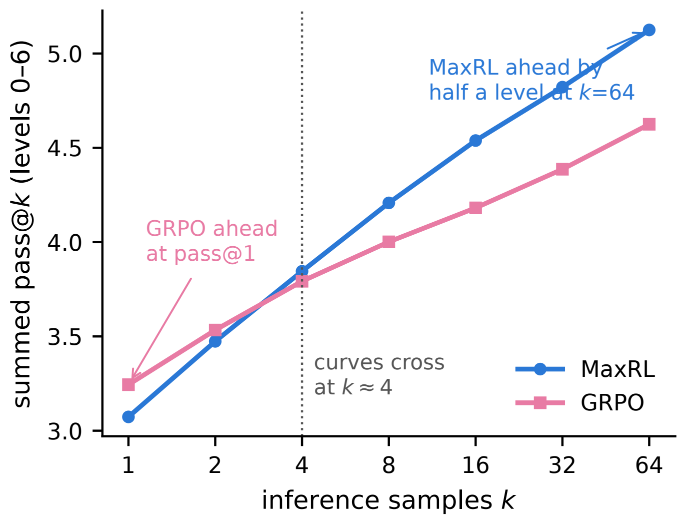

Figure 10: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** GRPO leads at pass@1; the curves cross at $k\approx 4$; MaxRL leads by half a level at $k=64$ — choose the estimator by how many samples you will draw at inference.

Relabeling in a PPO-family loop: the wiring contract. For adopters (the details a config cannot express). *Placement*: relabeling runs after reward computation and **before** `compute_log_prob` — old log-probs must be computed under the *rewritten* conditioning, making the relabeled group an on-policy ML term with the estimator's own weights, not a policy-gradient ratio against stale context (our gridworld ablation priced the stale-context mistake at −0.06 AUC, worse than no hindsight). *Rewrite*: the prompt's goal slot and the response's goal mentions are both rewritten (token IDs, attention mask, and position IDs rebuilt; 2-D RoPE only — 3-D mrope is refused loudly); the answer artifact is untouched by construction. *Group semantics*: each parseable failure relabels to its own achieved goal and keeps its group id, so one group can mix relabeled tasks; unparseable rollouts stay failures, preserving contrast ($K < N$). The group contrast is then "reached some verified goal" vs. "reached none" — coarser than 1-of-N of a single task, pre-registered as a deviation, with the verl-normalized advantage verified equal to $N\times$ the reference weights in direction; as Remark 3 states, this per-row variant is a weighted-SFT update, not a per-destination MaxRL gradient. *Hygiene*: the curriculum posterior observes pre-relabel scores only; no KL-to- reference is applied to relabeled groups in our runs (`kl_coef=0`). Reference implementation vendored in this repository (`verl_integration/vendored/hindsight.py`, ∼270 lines, with execution-commit provenance and CPU integration tests).

Jugs suite (pool-conditionality; §8). Tier grid t0–t4 (2–5 jugs, disjoint min-moves bands, 200 unique tasks/tier); feasibility landscape (SmolLM2-360M two-shot, N=16): t0 .058/.475 (pass@1/pass@16), t1 .006/.100, t2–t4 0/0 with relabel-yield-on-failure .62–.73 on every tier. All nine pre-registered B-arm cells (no-recycling / ungated / gated × 3 seeds) converged to the same deterministic fixed point: t0 mean@16 = pass@16 = 0.450 exactly (the 2-move template stratum), t1 pass@16 0.15 → 0, actor entropy 1.36 → 0.02–0.21. Ungated recycling ran 169 relabels/step at yield .88 and ended with the lowest entropy (0.02–0.03); the value-keyed gate rejected 99.8% of relabels (the granularity bug described in §8, since fixed and regression-tested). Postmortem and per-cell artifacts in the repository.

6.9 secondary reads (full). Pool construction and baseline calibration: the pre-SFT base model's relabel yield is 0.000 (n=384) — the syntactic prerequisite; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant ∼75% of the independence bound $1-(1-\text{pass@1})^{16}$ (ratios .744/.783/.755 across tiers), the baseline the headroom gate measures against. Training-side damage: ungated recycling ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired +.083/+.063/+.034). Entropy at 3 seeds: the gate's retention replicates (+.072 paired over baseline, all seeds positive, ∼1.6×);

ungated recycling's endpoint entropy is seed-unstable (spread ±.062, 4–9× the other arms'), consistent with its noisy frontier-coverage endpoints.

D Hyperparameters and compute

Total compute for every experiment in this paper: ∼230 A10G-hours on a single GPU plus CPU-days for the exact-gradient suites. Step-time telemetry (artifact `generation_timing.json`, 1,188 mixed GSM8K/Countdown steps): step-weighted mean generation fraction 31% overall; the eight generation-dominated sessions (>80% session mean, 282 steps) average 83% — a selected-subset description, not an estimate of typical cost. Table 2 gives the per-suite configuration and Table 3 the knob inventory with provenance.

Table 2: Per-suite configuration. Teacher knobs (decay, floor, γ) are shared across suites with the defaults below; what varies per suite is scale, not schedule.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
rollouts N	16	32	16	16
groups/step	1	8	64	64
optimizer, lr	SGD, 0.5	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s / 250 steps†	50 steps	60 steps
seeds (headline)	5–10	3 / 6 blocks†	1–2/cell	3/arm
compute/run	CPU-min	0.7 A10G-h	∼20 A10G-h	∼6 A10G-h

†cohort runs (§6.3a) used 2,400 s wall-clock; the balanced factorial and its confirmation wave use 250 fixed steps (matched generation and optimization budget) over six seed blocks — the two protocols are never mixed within a comparison.

Table 3: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
peak location/N-scaling	—	**derived** (Prop. 1)	the theory claim
posterior decay	0.7	tuned (V2b)	closes ∼19% of oracle gap
uniform floor	0.1	tuned	0.15 on maze; robust 0.05–0.2
γ (concentration)	1	tuned (V6)	4 on chains; no transfer to maze
`gate_max_p`	0.5	tuned (p=1 zero vacuous)	one trade point validated (§6.9)
relabel dose cap	12 groups/step	tuned	sets *where* coverage is spent (§6.4)

Reproducibility

Every proposition has an enumeration/Monte-Carlo verification script (`test_mass_formulas.py` in `curriculum_maxrl/` checks the MaxRL, RLOO, and both GRPO-SD mass formulas against exact enumeration and the deployed code). Every figure regenerates from committed inputs listed, with checksums and commands, in the frozen manifest `paper/results/ manifest.json`: result values live in versioned JSON tables under `paper/figures/data/` (none are hard-coded in figure scripts), and the maze cohort of Fig. 5 is frozen in an explicit run manifest — adding a run file cannot silently change the figure, and missing inputs are a hard error. The Jugs preregistration, raw cells, verdicts, and postmortem are vendored in `jugs_llm/results/` with preregistration- and execution-commit provenance. The LLM-scale relabeling implementation is vendored verbatim with execution-commit provenance (`verl_integration/vendored/`), so the loop the Countdown and Jugs suites executed is inspectable in this repository. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team's summary (raw logs live with that team); per-step timing fractions come from verl's in-run telemetry (`generation_timing.json`); the fig2/fig3 endpoint tables were transcribed from run analyses, and all of fig2's quantitative panels now carry checked-in derivation scripts that regenerate them from per-seed artifacts and fail on mismatch (`verify_fig2a_from_artifacts.py` covers panels a–b, `verify_fig2c_from_logs.py` panel c against raw maze seed logs) — only the GSM8K panel (d) and fig3 remain transcribed, with per-panel provenance in the data files; and the GSM8K pass@k harness sits $\sim 3\times$ below the trainer's val metric (unreconciled; only within-sweep contrasts are quoted, and the reconciliation script is queued behind the running arms). A one-row-per-run registry (`curriculum_maxrl/run_registry.json`; 48 rows spanning the maze, Countdown, and GSM8K artifacts, with arm/seed/protocol per row) indexes every run this paper draws on; the maze main-effect subset and its 5v4 composition are tabulated in `permutation_reanalysis.json`, committed alongside. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.